Wenyue Zou

Wenyue Zou | Ph.D. in Environmental Science
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Expertise Centre for Climate Extremes (ECCE)
Faculty of Geosciences and Environment, University of Lausanne
Chair of Hydrology and Resources Management
Institute of Environmental Engineering, ETH Zurich

RESEARCH INTERESTS

- Storm-flood hazards and risk assessment
- Statistical/ML and dynamic modelling of extreme rainfall
- Impact of climate extremes

PROFESSIONAL SKILLS

- Extreme rainfall frequency analysis, spatial downscaling, and stochastic storm modeling
- Future storm projection and their effect on urban flood response
- MATLAB, Python, and ArcGIS for mapping and data analysis
- Project design, data management, and interdisciplinary collaboration

PUBLICATIONS

[5] Zou, W et al., 2025. A process-based framework linking temperature-rainfall projections and urban flood modeling. *Hydrology and Earth System Sciences*, under review. https://doi.org/10.5194/egusphere-2025-4099.

[4] Zou, W., Wright, B., Peleg, N. 2025. Morphing sub-daily rainfall fields based on temperature shifts to project future changes in rainfall extremes. *Water Resources Research*, https://doi.org/10.1029/2024WR038847.

[3] Zou, W et al., 2024. Multiple-point geostatistics-based spatial downscaling of heavy rainfall fields. Journal of Hydrology. https://doi.org/10.1016/j.jhydrol.2024.130899.

[2] Zou, W et al., 2021. Spatial interpolation of the extreme hourly precipitation at different return levels in the Haihe River basin. *Journal of Hydrology 598, 126273*. https://doi.org/10.1016/j.jhydrol.2021.126273.

[1] Li, Q., Zhou, J., **Zou, W**., et al., 2020. A tributary-comparison method to quantify the human influence on hydrological drought. *Journal of Hydrology*. https://doi.org/10.1016/j.jhydrol.2020.125652.

Manuscripts in review:

[8] Zhu, Y., Burlando, P., Zhang, Y., Wang, J., **Zou, W**., Fatichi, S., 2025. Influence of Urban Landscape Morphology Changes During Urban Expansion on Pluvial Flooding. *Sustainable Cities and Society*.

[7] Li, T., Yin, S., Xiao, Y., Wang, M., **Zou, W**., Peleg, N., 2025. Quantifying rainfall variability and potential hazards of extreme events in Beijing through stochastic simulations. *Journal of Hydrology*.

[6] Nakhael, N., Li, R., Chen,Y., **Zou, W**., Ni, G., 2025. Scalable Urban Pluvial Flood Prediction with Tile-Based Super-Resolution Modeling. *Water Research*.

Main achievements

Generating high-resolution gridded rainfall datasets and extreme value analysis

During my master's and the first year of my PhD, I led two projects to generate high-resolution rainfall extreme datasets from gauged observations and coarse satellite data, with all datasets and methods open-sourced for the research community.

Context and challenge: High-resolution spatial rainfall extremes (e.g., 1 km, hourly) are required for hydrological applications in mountainous and urban catchments. However, most basins rely on sparse gauge networks, requiring robust spatial interpolation to fill ungauged locations. Satellite products can provide spatial rainfall, but their resolution (~10 km) is too coarse to use and needs to be downscaled not only in rainfall intensity but also in spatial structure.

To meet this challenge, I evaluated six different interpolation methods and identified that the Kriging with external drift (KED) method outperformed the others for interpolating hourly station extreme rainfall, generating 2- to 100-year return level maps across a major Chinese basin. This method contributes to more accurate estimation of spatial rainfall extremes and can also be applied in other regions [2].

The second way to meet this challenge is downscaling satellite products: I designed a Multiple-point geostatistics-based spatial (MPS) downscaling model, which learns storm patterns from a few years of radar data, to downscale long-term satellite rainfall into high-resolution. I applied it to successfully downscale 22 years of satellite rainfall from 8 to 1 km over Beijing city. The results indicate that the model downscales well in both rainfall intensity and spatial structure. The proposed approach can be applied to other rainfall datasets and regions, enhancing the availability of fine-scale rainfall datasets for diverse hydrometeorological applications [1,3].

These open-source, scalable solutions provide critical tools for generating high-resolution rainfall datasets and accurate spatially distributed rainfall extreme estimates for hydrological design in data-scarce regions. The resulting datasets and methods support advanced hydrometeorological risk assessment and applications [1,2,3].

Projecting future changes in storms and urban flood hazards under climate warming

During my PhD, I led projects on future changes in rainfall extremes and their effect on urban flooding with climate warming, developing and coupling a temperature-rainfall projection method, a stochastic storm transposition (SST), and a hydrodynamic flood model.

Context and challenge: Predicting future shifts in storms and urban pluvial floods is vital for disaster prevention. The rapid hydrological response and highly variable storm fields demand high-resolution rainfall data and flood simulation for robust scenario analysis. However, future sub-daily rainfall is not readily available globally due to the high computational cost of climate models.

To meet this challenge, I introduced a Distributed-based spatial quantile mapping (DSQM) method that uses rainfall-temperature scaling to project future rainfall fields under different warming scenarios. By integrating it with SST, I efficiently estimated future rainfall extremes spatially. I demonstrated the efficiency of the DSQM-SST approach by projecting changes in Beijing's hourly rainfall extremes under multiple regional warming levels [4].

I subsequently simulated the DSQM-SST-derived storms into a high-resolution hydrodynamic flood model, to generate future urban flood depth, velocity and inundation. The simulations revealed a nonlinear increase in flood hazard with intensified local storms. The proposed DSQM-SST-Flood framework offers a data-driven, efficient, and physically grounded approach for assessing future urban storms and flood risk under climate warming. It bridges the gap between the limitations of high-resolution rainfall projections and practical engineering needs, providing actionable outputs for climate-resilient infrastructure design and adaptation planning [5].

Engaging the research community and broader society to amplify impact

I am committed to ensuring my research creates tangible value beyond academia by supporting the scientific community, mentoring students, and translating knowledge for public benefit. My contributions span three key areas:

1. Strengthen scientific rigor by providing rigorous peer review for leading journals (e.g., *Earth System Science Data*), enhancing the quality and clarity of publications in hydrometeorology and climate science.

2. Foster collaborative climate actions by serving on organizing committees for two university-based centers (UNIL's Expertise Center for Climate Extremes and UNIL-EPFL Center for Climate Impact and Action), helping design and organize workshops to raise awareness of climate-change impacts and build cross-institutional partnerships to advance solutions.

3. Mentor students as a teaching assistant on key practical courses (e.g., Climate Change Impact Assessment and Watershed Modeling), providing hands-on training in data collection, numerical modeling, and climate impact analysis.

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EDUCATION BACKGROUND

- **2021 - present** **Ph.D. in Environmental Science,** Faculty of Geosciences and Environment, University of Lausanne, Switzerland

  Thesis: ***Future changes in rainfall properties and their effect on urban flooding***
  Supervisor: Nadav Peleg

- **2018 - 2021** **M.sc in Physical Geography,** Faculty of Geographic Science, Beijing Normal University, China GPA:3.75/4 (10%)

  Thesis: ***Spatiotemporal characteristics of rainfall events based on a highly dense rain-gauge network.*** *Supervisor: Shuiqing Yin*

- **2014 - 2018** **B.sc in Geography,** Faculty of Geographic Science, Northwest Normal University, China GPA:3.84/4 (1%)

  Thesis: ***Spatial and temporal variation characteristics of hourly precipitation during the Warm season 1961-2012 in the Haihe River basin.***
  Supervisor: Junju Zhou

SERVICES

- 2024 **Committee of Expertise Center for Climate Extremes, University of Lausanne**
- 2022 **Teaching assistant,** Faculty of Geosciences and Environment, University of Lausanne
  - Watershed and river network modelling
- 2020 **Teaching assistant,** Faculty of Geographical Science, Beijing Normal University
  - Meteorology and Climate Practice course
  - Assessment of climate change and its impacts
- 2023 Student committee of CLIMACT between UNIL and EPFL university
- 2021 Student committee in Association du Corps Intermédiare, FGSE, UNIL

AWARDS AND SCHOLARSHIPS

- 2025 Swiss-China Innovation & Entrepreneurship International Competition, Global Finalist (15%)
- 2021 Excellent Master Thesis in Beijing Normal University
- 2018, 2019 Academic Scholarship at Beijing Normal University (First class, twice, 10%)
- 2017 First prize in Scientific Research Challenge Cup at Northwest Normal University (1%)
- 2016 First prize in National College Students Mathematic Modeling Competition, China (10%)
- 2015 National Endeavor Scholarship, China (10%)